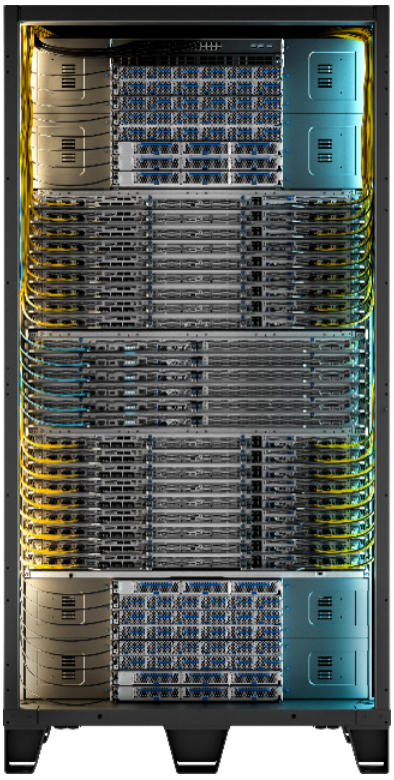
Supermicro AMD Helios Platform

An Open Rackscale AI Infrastructure for Inference, Training, and Fine-Tuning



The Supermicro AMD Helios platform, AMD's rack-scale AI infrastructure (a double-wide OCP Open Rack Wide platform), is built around the next-generation AMD Instinct™ MI455X accelerators. These are data-center AI/HPC GPUs based on the advanced CDNA™ architecture. The MI455X is the flagship GPU powering Helios racks, designed for massive-scale AI training, inference, and trillion-parameter models.

Key Features:

- 18 6th Gen AMD EPYC™ processors
- 72 AMD Instinct MI455X GPUs
- FP4 FLOPS: 2.9 exaFLOPs; FP8 FLOPS: 1.4 exaFLOPs
- HBM4 Memory Capacity: Up to 31 TB
- Memory Bandwidth: Up to 1.4 PB/s
- Scale-up Bandwidth: Up to 260 TB/s
- Scale-out Bandwidth: Up to 43 TB/s

AMD Instinct MI455X GPUs

Built on AMD CDNA architecture, the Supermicro AMD Helios platform combines massive HBM4 capacity, extreme memory bandwidth, and multi-petaflop compute performance to accelerate large-scale inference and AI training across the Helios rack-scale platform. The MI455X GPUs enable efficient scaling for large models and high-throughput AI infrastructure deployments.

6th Gen AMD EPYC CPUs

The new AMD CPUs deliver high core counts, fast CPU-to-GPU connectivity, and high-bandwidth memory access to support AI orchestration, training, and inference workloads at rack scale. These CPUS are optimized for modern AI infrastructure, EPYC processors help orchestrate networking, storage, and workloads at rack scale.

AMD Pensando™ "Vulcano" 800 AI NICs

Delivers leadership performance with 800 Gbps Ethernet speeds is the only AI NIC to offer up to 2.4 Tbps of scale-out bandwidth per GPU, with full hardware and software programmability to keep GPUs and CPUs continuously fed at rack scale.

Scale-up Fabric

Built on an open networking approach, the Helios platform leverages UALink™ over Ethernet (UALoE) to deliver high-bandwidth, low-latency GPU connectivity at rack scale. Powered by advanced Ethernet silicon, it enables seamless interconnect up to 72 GPUs per Helios rack, supporting efficient scaling and reduced vendor lock-in.



AMD Instinct MI455X GPUs Spec Details

- Up to 432 GB of HBM4 memory
- Up to 19.6 TB/s peak memory bandwidth
- 10x Gen vs Gen performance

6th Gen AMD EPYC Processors Spec Details

- 2.0x CPU to GPU bandwidth
- 1.7x Gen vs. Gen performance
- 1.6 TB/s memory bandwidth

AMD Pensado™ "Vulcano" 800 AI NIC Spec Details

- Up to 800 Gbps high-performance Ethernet network throughput
- Up to 8× scale-out bandwidth per GPU
- UAL | PCIe 6.0 host interface for low-latency GPU communication
- UEC-ready RDMA Ethernet optimized for large-scale AI clusters

AMD Pensado™ Salina DPU Spec Details

- Up to 400 Gbps throughput
- Up to 2x performance and scale over previous generation DPUs
- 18 DPUs per rack

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